

(% i4)

```
/* Declare variables*/  
declare(x, real);  
declare(k, real);  
declare(E, real);  
assume(x > 0, x < 1, E < 0, k >= - %pi, k <= %pi);
```

done

done

done

$$[x > 0, x < 1, E < 0, k \geq -\pi, k \leq \pi]$$

(% i5)

```
/* Solve the differential equation*/  
sol: desolve(- diff(u(x), x, 2) - 2 * %i * k * diff(u(x), x, 1) + (k^2 - E) * u(x) = 0, u(x));
```

$$u(x) = e^{-i k x} \left(\frac{\sinh(\sqrt{-E} x) \left(2 \left(\frac{d}{dx} u(x) \right)_{x=0} + 2 i u(0) k \right) - 2 i u(0) k}{2 \sqrt{-E}} + u(0) \cosh(\sqrt{-E} x) \right)$$

(% i6)

```
/* Rewrite conditions with C_1 and C_2*/  
sol: rhs(subst([  
    at(diff(u(x), x, 1), x = 0) = C_2,  
    at(u(x), x = 0) = C_1], sol));
```

$$e^{-i k x} \left(\frac{\left(2 (2 i C_1 k + C_2) - 2 i C_1 k \right) \sinh(\sqrt{-E} x)}{2 \sqrt{-E}} + C_1 \cosh(\sqrt{-E} x) \right)$$

(% i7)

```
/* Periodic condition*/  
eq1: factor(at(sol, x = 0) = at(sol, x = 1));
```

$$C_1 = \frac{e^{-i k} \left(i C_1 \sinh(\sqrt{-E}) k + C_1 \cosh(\sqrt{-E}) \sqrt{-E} + C_2 \sinh(\sqrt{-E}) \right)}{\sqrt{-E}}$$

(% i9)

```
declare(g, real);  
assume(g > 0);
```

done

$$[g>0)$$

(% i10)

/* Delta jump condition*/

eq2: factor(at(diff(sol, x, 1), x = 1) = at(diff(sol, x, 1), x = 0) + g * at(sol, x = 0));

$$(\% e^{-(\% i k)} (C_1 \sinh(\sqrt{-E}) k^2 - \% i C_2 \sinh(\sqrt{-E}) k - C_1 \sinh(\sqrt{-E}) E + C_2 \cosh(\sqrt{-E}) \sqrt{-E})) / \sqrt{-E} = C_1 g + C_2$$

(% i11)

/* Matrix form*/

A: coefmatrix([eq1, eq2], [C_1, C_2]);

$$\begin{bmatrix} -\left(\frac{\% e^{-(\% i k)} (\% i \sinh(\sqrt{-E}) k + (\cosh(\sqrt{-E}) - \% e^{\% i k}) \sqrt{-E})}{\sqrt{-E}}\right) & -\left(\frac{\% e^{-(\% i k)} \sinh(\sqrt{-E})}{\sqrt{-E}}\right) \\ \frac{\% e^{-(\% i k)} (\sinh(\sqrt{-E}) k^2 - \% e^{\% i k} \sqrt{-E} g - \sinh(\sqrt{-E}) E)}{\sqrt{-E}} & -\left(\frac{\% e^{-(\% i k)} (\% i \sinh(\sqrt{-E}) k + (\% e^{\% i k} - \cosh(\sqrt{-E})) \sqrt{-E})}{\sqrt{-E}}\right) \end{bmatrix}$$

(% i12)

/* Determinant of A*/

d: trigsimp(determinant(A));

$$(\% e^{-(2 \% i k)} (\% e^{\% i k} \sinh(\sqrt{-E}) \sqrt{-E} g + (2 \% e^{\% i k} \cosh(\sqrt{-E}) - \% e^{2 \% i k} - 1) E)) / E$$

(% i14)

declare(s, real);

assume(s > 0);

done

$$[s>0)$$

(% i15)

/* Let varepsilon = - s^2 */

d: subst(E = - s^2, d);

$$-\left(\frac{\% e^{-(2 \% i k)} (\% e^{\% i k} g s \sinh(s) - s^2 (2 \% e^{\% i k} \cosh(s) - \% e^{2 \% i k} - 1))}{s^2}\right)$$

(% i17)

/* Band structure*/

eq: solve(d = 0, cosh(s));

$$\left[\cosh(s)=\frac{e^{-ik}\left(e^{ik}g\sinh(s)+\left(e^{2ik}+1\right)s\right)}{2s}\right]$$